Acceleration, signs and turning around

LEARNING TARGET · *R*4 I can find acceleration from changes in velocity.

A · KEY VOCABULARY

Average a	Change in velocity divided by elapsed time.
Speeding up	v and a have the same direction in 1D.
Slowing down	v and a oppose each other before the turn.
Turn	$v = 0$ for an instant; a can be nonzero.

B · KEY EQUATIONS + UNITS

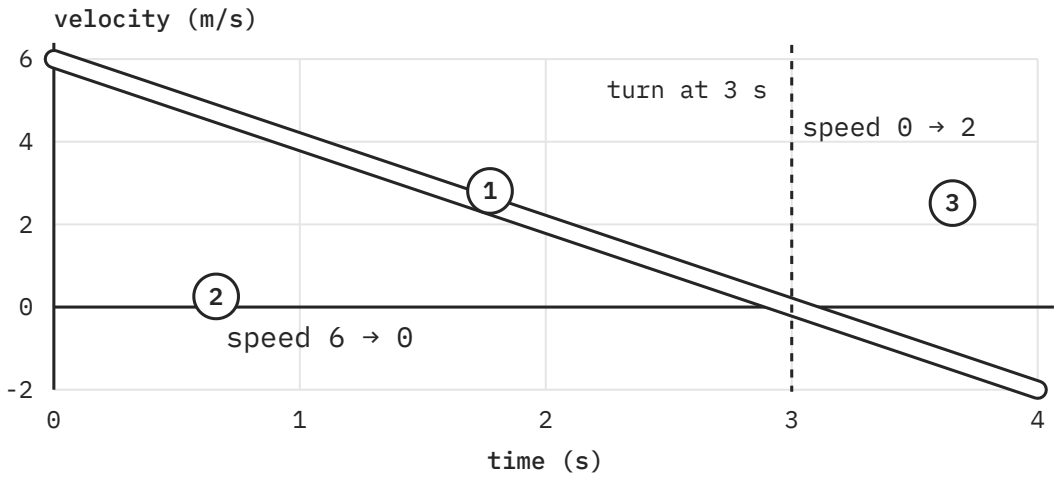
$$\bar{a} = \Delta v \div \Delta t$$

$$v_f = v_i + at$$

a	m/s²
v rule	constant a only

C · COLOR IT · SLOW, THEN SPEED UP

Right is +; constant a.



☐ **1 · PURPLE** Signed velocity

☐ **2 · RED** Before turn: slowing

☐ **3 · GREEN** After turn: speeding up

D · WORKED EXAMPLE

A cart's velocity changes uniformly from **+6 m/s** to **−2 m/s** in **4 s**. **Find its acceleration and the time it reverses.**

GIVEN	EQUATION	WORK	ANSWER
$v_i = +\mathbf{6}$ **m/s** $v_f = -\mathbf{2}$ **m/s** $\Delta t = \mathbf{4}$ **s**	$a = \Delta v/\Delta t$. Turn: $0 = v_i + at$.	$a = (-2 - 6)/4$ $0 = 6 - 2t$ $t = \mathbf{3}$ **s**	$a = -\mathbf{2}$ **m/s²** Slows for **3 s**, then speeds up to the left.

E · THE 2026-XJ THREAD

DAY 7 OF 22 · NEW TOOL: SIGNED CHANGE

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else